

Report

Date	10 Oct 2025
Team ID	xxxxxx
Project Name	Predicting Plant Growth Stages with Environmental and Management Data Using Power BI
Maximum Marks	5 Marks

Overview

This report presents an analytical study of **plant growth patterns** using environmental and soil data. The dataset, collected from Kaggle's *Plant Growth Data Classification*, was visualized in Power BI to extract meaningful insights about how different environmental factors such as temperature, humidity, and soil type influence plant growth stages.

The purpose of the report is to provide a **data-driven understanding** of plant development trends, identify the most influential growth factors, and support optimized agricultural management strategies.

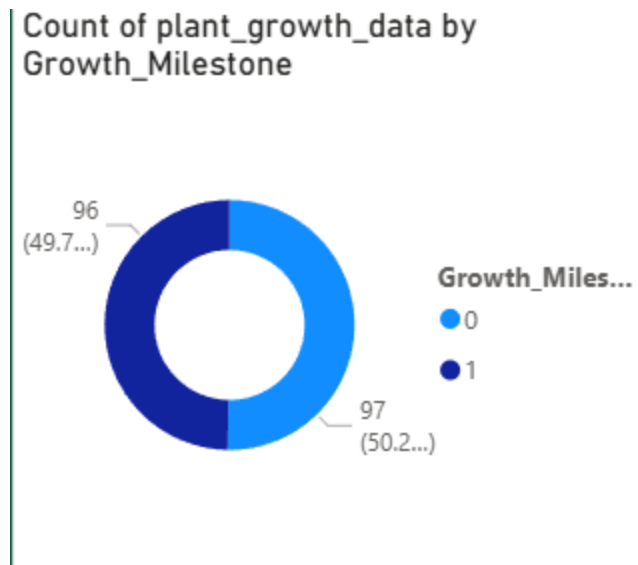
Report Objectives

1. Analyze and visualize relationships between environmental variables and plant growth stages.
2. Identify key conditions (soil, temperature, humidity) that promote healthy growth.
3. Detect trends and correlations that can inform smart farming decisions.
4. Develop an interactive Power BI report for effective monitoring and prediction.

Key Visualizations and Observations

1. Growth Stage Distribution

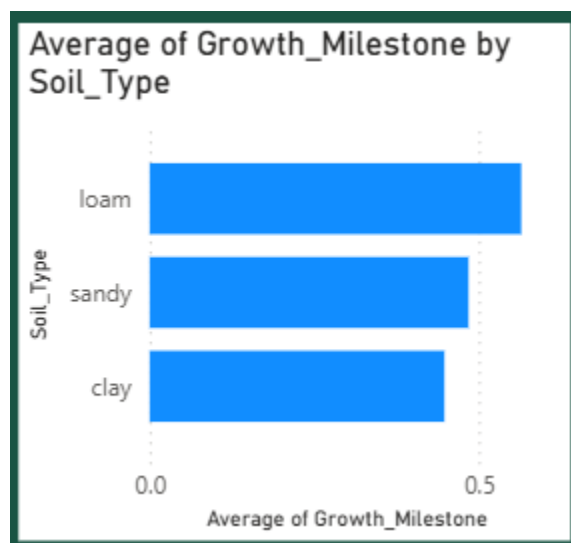
- **Visualization:**



- **Insight:** Indicates the dataset is rich in early-stage growth data, useful for early growth analysis.

2. Average Growth by Soil Type

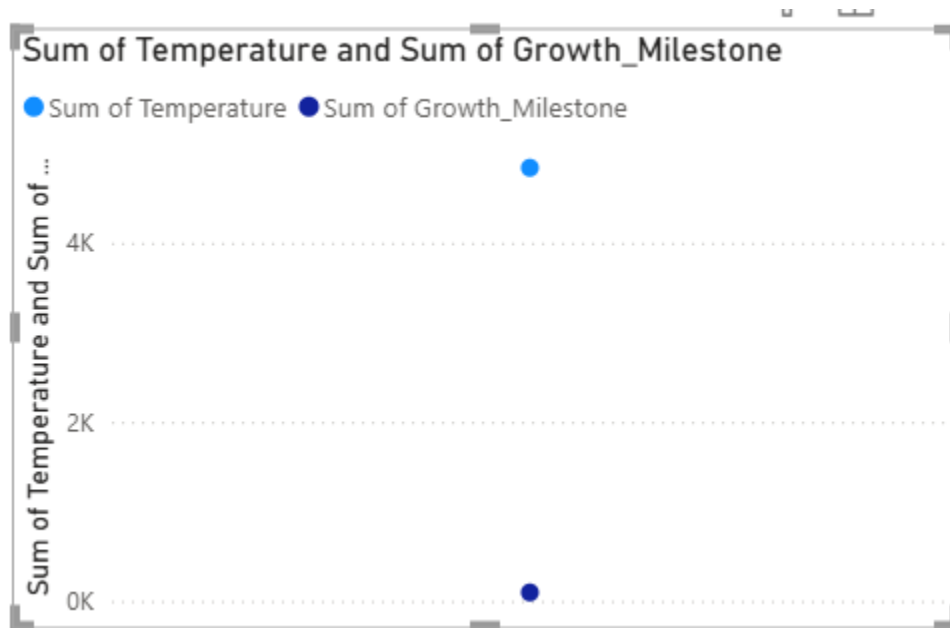
- **Visualization:**



- **Observation:** Loamy soil shows the highest average growth milestone, while Sandy soil records the lowest.
- **Insight:** Loamy soil's balanced texture retains moisture while allowing aeration, leading to better growth.

3. Impact of Temperature on Growth

- **Visualization:**



- **Insight:** Plants exhibit optimal growth within moderate temperature ranges, suggesting heat stress beyond threshold limits.

4. Humidity Influence on Plant Growth

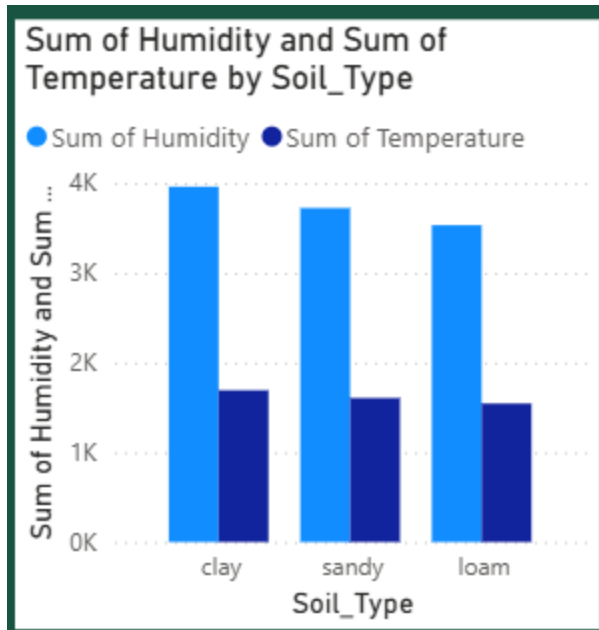
- **Visualization:** Scatter Plot
- **Observation:** Moderate humidity (60–70%) correlates with the highest growth milestone scores.
- **Insight:** Consistent humidity plays a vital role in maintaining healthy plant development.

5. Average Growth by Region / Location

- **Visualization:** Map Chart
- **Observation:** Regions with stable temperature and humidity show higher growth performance.
- **Insight:** Regional variation suggests environmental conditions and soil fertility differences.

6. Combined Factor Analysis

- **Visualization:**



- **Observation:** Mature stage plants require slightly higher humidity and moderate temperatures compared to seedlings.
- **Insight:** Environmental control can be optimized differently for each growth stage.

Final:

- Loamy soil provides **best growth conditions** among all soil types.
- Optimal temperature range: **20°C–30°C**.
- Humidity between **60–70%** supports steady growth.
- **Temperature** is the strongest predictor of plant growth stage.
- **Regional and seasonal** variations affect growth performance.
- The **data-driven approach** helps optimize irrigation and fertilization strategies.

Report Design Highlights

- Clear and interactive Power BI dashboards with filters for:
 - Soil Type
 - Temperature Range

- Growth Stage
 - Region
- Drill-down features for exploring data by location and growth stage.
- Use of **color-coded heatmaps** for quick correlation understanding.
- Consistent layout using green, brown, and beige tones to reflect agricultural context.